Exercise 7.1 *Suppose that in the definition of the $(2^{nR}, n)$ code for the DMC $p(y|x)$, we allow the encoder and the decoder to use random mappings. Specifically, let K be an arbitrary random variable shared by sender and receiver and independent of the message M and the channel. The encoder generates a codeword $x^n(m,k)$, and the decoder generates an estimate $\hat{m}(y^n,k)$ depending on the value of k. Show that this randomization does not increase the capacity of the DMC.*
Hint: *Write $H(M) = I(M : Y^n, W) + H(M|Y^n, W)$, and use Fano's inequality.*

Solution: Note that

$$nR = H(M) = I(M : Y^n, W) + H(M|Y^n, W).$$

By Fano's inequality, $H(M|Y^n, W) \leq n\epsilon_n$ for some $\epsilon_n \to 0$ as $n \to \infty$. Thus,

$$\begin{aligned}
nR &\leq I(M : Y^n, W) + n\epsilon_n \\
\implies n(R - \epsilon_n) &\leq I(M : W) + I(M : Y^n|W) \\
&= I(M : Y^n|W) \\
&= \sum_{i=1}^{n} I(M : Y_i|Y^{i-1}, W) \\
&\leq \sum_{i=1}^{n} I(M, Y^{i-1}, W : Y_i) \\
&\leq \sum_{i=1}^{n} I(M, Y^{i-1}, W, X^i : Y_i) \\
&= \sum_{i=1}^{n} I(X_i : Y_i),
\end{aligned}$$

where the last equality follows from the fact that $p(y_i|x^i, y^{i-1}, m, w) = p_{Y|X}(y_i|x_i)$. Finally, we have

$$R - \epsilon_n \leq \frac{1}{n} \sum_{i=1}^{n} I(X_i : Y_i) \leq \max_{P_X} I(X : Y) = C,$$

and the claim is proven.

Exercise 7.2 *Consider that there is a binary symmetric channel (BSC) with crossover probability ϵ. We use a coding scheme on this channel that encodes messages a_1 and a_2 as 000 and 111, respectively. On the other hand, we use a decoding scheme that relies on the majority principle, i.e., 001 $\mapsto$ 0 and 011 $\mapsto$ 1.*

a) *Compose the above encoder, the BSC, and the above decoder into a new channel. Describe the input/output alphabets and the channel rule (the conditional probability of outputs given inputs) of this new channel.*

b) *Assume the original BSC has crossover probability $\epsilon = 0.1$. What is its channel mutual information? Compare it to the channel mutual information of the composed channel.*

c) *Suppose that we have 4 input messages (each to be sent with equal probability). Find a code of length 3 that minimises the average decoding error, and compute it.*

d*)* *Consider a binary erasure channel (BEC) with erasure probability* $\epsilon = 0.1$. *Suppose we encode the messages* a_1 *and* a_2 *as* 000 *and* 111, *respectively. On the decoder side, if we receive* $\perp\perp\perp$, *we will randomly assign it to* a_1 *or* a_2 *with equal probability; otherwise, we decode according to the surviving symbol, i.e.,* $0\perp\perp\mapsto a_1$ *and* $\perp\perp 1 \mapsto a_2$. *What is the average decoding error in this case?*

Solution:

a.) Both the input and output alphabets are still $\{0,1\}$. Denoting the original BSC by W, and the new channel by $\tilde{W}$, we have

$$\begin{aligned}\tilde{W}(0|0) &= W^{\otimes 3}(000|000) + W^{\otimes 3}(001|000) + W^{\otimes 3}(010|000) + W^{\otimes 3}(100|000)\\ &= (1-\epsilon)^3 + 3\epsilon(1-\epsilon)^2 = 1 - 3\epsilon^2 + 2\epsilon^3,\\ \tilde{W}(1|0) &= 1 - \tilde{W}(0|0) = 3\epsilon^2 - 2\epsilon^3,\end{aligned}$$

and by symmetry $\tilde{W}(1|1) = 1 - 3\epsilon^2 + 2\epsilon^3$ and $\tilde{W}(0|1) = 3\epsilon^2 - 2\epsilon^3$. In other words, the resultant channel is a BSC with crossover probability $3\epsilon^2 - 2\epsilon^3$.

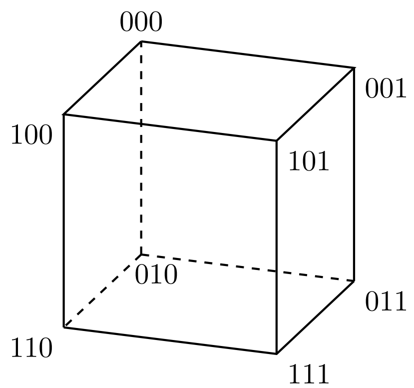
b.) Let us denote $h(\epsilon) := \epsilon \log \frac{1}{\epsilon} + (1-\epsilon)\log\frac{1}{1-\epsilon}$. The crossover probability of the new channel is $\hat{\epsilon} = 0.028$. The channel capacity per channel use is given by

$$\begin{aligned}C &= \max I(\hat{M} : M)\\ &= \max H(\hat{M}) - \sum_{m\in\{a_1,a_2\}} P_M(m) H(\hat{M}|M=m)\\ &= \max H(\hat{M}) - h(\hat{\epsilon})\\ &= 1 - h(0.028)\\ &= 0.81574 \text{ bits per channel use.}\end{aligned}$$

The code rate is $\frac{1}{3}$. Thus, the channel capacity in bits per transmission is $\frac{C}{3} = 0.27191$ bits per transmission.

The capacity of the original channel is simply $1 - h(0.1) = 0.531004$.

c.) To find the best codewords of four messages with length 3, we should let the codewords be as far as possible.



From the codewords cube, we can see that one possible optimal set is $\{000, 101, 011, 110\}$. Each of these codewords has two different bits comparing to other codewords. The decoding rule is to assign received sequences to the nearest codewords. If we have exactly one error, there are three nearest codewords and randomly choosing one decodes correctly with probability $\frac{1}{3}$. Hence,

$$P(\text{recovered} = \text{transmitted}) = 0.9 \times 0.9 \times 0.9 + \frac{1}{3} \cdot 3 \cdot 0.9 \times 0.9 \times 0.1 = 0.81.$$

The probability of error is 0.19.

d.) The received sequences $00\perp$, $0\perp 0$, $\perp 00$, $\perp\perp 0$, $\perp 0\perp$, $0\perp\perp$, 000 can be decoded as 0. The received sequences $11\perp$, $1\perp 1$, $\perp 11$, $\perp\perp 1$, $\perp 1\perp$, $1\perp\perp$, 111 can be decoded as 1. Only the $\perp\perp\perp$ may cause the error. The probability of receiving $\perp\perp\perp$ is 0.1^3. Thus, the probability of error for this code is $0.1^3/2 = 0.0005$.

Exercise 7.3 (Feinstein's bound) *Let $W_{Y|X}$ be a channel from input set $\mathcal{X}$ to output set $\mathcal{Y}$.*

a) *Suppose we use the channel* once. *Show that there exists a code with M codewords with average probability of error ϵ satisfying*

$$\epsilon \leq P\left[\log \frac{W_{Y|X}(Y|X)}{P_Y(Y)} \leq \log M + \gamma\right] + 2^{-\gamma}.$$

for any choice of $\gamma > 0$ and any input distribution P_X where $P_Y(y) = \sum_x W_{Y|X}(y|x)P_X(x)$. A stronger version of this bound (for maximum error) was shown by Feinstein.
Hint: *Generate codewords independently according to P_X. Instead of using typical set for decoding, use $\hat{m} \in \{1, \ldots, M\}$ as the transmitted message if it is the unique one satisfying*

$$\log \frac{W_{Y|X}(y|x(\hat{m}))}{P_Y(y)} \geq \log M + \gamma.$$

If there is no unique $\hat{m}$ satisfying the above condition, declare an error.

b) *Use a) to prove achievability of the channel coding theorem for DMCs.*

c) *Again consider the setup in a). Let*

$$V := \mathrm{Var}\left[\log \frac{P^*_{Y|X}(Y|X)}{P^*_Y(Y)}\right]$$

be evaluated at a capacity-achieving input distribution P^*_X. Show that, by the central limit theorem, there exists a sequence of codes indexed by blocklength n, with sizes M_n satisfying

$$\log M_n = nC + \sqrt{nV}\Phi^{-1}(\epsilon) + o(\sqrt{n})$$

such that the average error probability is no larger than $\epsilon + o(1)$.

Solution:

a.) As provided in the hint, we generate M codewords $x(m)$ independently from P_X. To send message m, we transmit codeword $x(m)$. Decode using the rule given above. Without loss of generality, we assume $m = 1$ was sent. An error occurs if and only if one or more of the following events occurs:

$$\mathcal{E}_1 := \left\{\log \frac{P_{Y|X}(Y|X(1))}{P_Y(Y)} < \log M + \gamma\right\}$$
$$\mathcal{E}_2 := \left\{\exists\, \tilde{m} \neq 1 : \log \frac{P_{Y|X}(Y|X(\tilde{m}))}{P_Y(Y)} \geq \log M + \gamma\right\}$$

The probability of error can be bounded as

$$P(\mathcal{E}) \leq P(\mathcal{E}_1) + P(\mathcal{E}_2)$$

Now note that $(X(1), Y) \sim P_X P_{Y|X}$, and thus $P(\mathcal{E}_1)$ gives the first term in the bound we have to show. We simply have to show that $P(\mathcal{E}_2) \leq 2^{-\gamma}$. For this, consider

$$\begin{aligned}
P(\mathcal{E}_2) &= P\left(\exists\, \tilde{m} \neq 1 : \log \frac{P_{Y|X}(Y|X(\tilde{m}))}{P_Y(Y)} \geq \log M + \gamma\right) \\
&\overset{(a)}{\leq} \sum_{\tilde{m}=2}^{M} P\left(\log \frac{P_{Y|X}(Y|X(\tilde{m}))}{P_Y(Y)} \geq \log M + \gamma\right) \\
&\overset{(b)}{=} \sum_{\tilde{m}=2}^{M} \sum_{x,y} P_X(x) P_Y(y) \mathbf{1}\left\{\log \frac{P_{Y|X}(y|x)}{P_Y(y)} \geq \log M + \gamma\right\} \\
&\overset{(c)}{\leq} \sum_{\tilde{m}=2}^{M} \sum_{x,y} P_X(x) P_{Y|X}(y|x) M^{-1} 2^{-\gamma} \mathbf{1}\left\{\log \frac{P_{Y|X}(y|x)}{P_Y(y)} \geq \log M + \gamma\right\} \\
&\overset{(d)}{\leq} \sum_{\tilde{m}=2}^{M} \sum_{x,y} P_X(x) P_{Y|X}(y|x) M^{-1} 2^{-\gamma} \overset{(e)}{\leq} 2^{-\gamma}
\end{aligned}$$

where (a) is due to the union bound, and (b) is due to the fact that for $\tilde{m} \neq 1$, the codeword $X(\tilde{m})$ and channel output Y are independent, and (c) is due to the fact that we're only summing over all (x, y) such that $\log \frac{P_{Y|X}(y|x)}{P_Y(y)} \geq \log M + \gamma$, and for (d) we dropped the indicator, and for (e) we use the fact that $\sum_{x,y} P_X(x) P_{Y|X}(y|x) = 1$ and that there are $M - 1$ terms in the outer sum.

b.) We want to show that we can achieve a code rate arbitrarily close to C. Going to the n-fold (n channel uses setting), we have that there exists a code with blocklength n, M_n codewords and average error probability ϵ_n satisfying

$$\epsilon_n \leq P\left(\log \frac{P_{Y^n|X^n}(Y^n|X^n)}{P_{Y^n}(Y^n)} \leq \log M_n + \gamma\right) + 2^{-\gamma}.$$

Choose P_{X^n} to be the n-fold product distribution corresponding to a capacity-achieving input distribution $P_X \in \arg\max_{P_X} I(X;Y)$. We now set $M_n = 2^{n(C-2\gamma')}$. (As we will see next, any $M_n = 2^{n(C-c\gamma')}$, with a constant $c > 1$ would work. Importantly, larger values for M_n will fail.)
Since the channel is a DMC, we have

$$\begin{aligned}
&P\left(\log \frac{P_{Y^n|X^n}(Y^n|X^n)}{P_{Y^n}(Y^n)} \leq \log M_n + \gamma\right) \\
&= P\left(\sum_{i=1}^{n} \log \frac{P_{Y|X}(Y_i|X_i)}{P_Y(Y_i)} \leq \log M_n + n\gamma'\right) \\
&= P\left(\sum_{i=1}^{n} \log \frac{P_{Y|X}(Y_i|X_i)}{P_Y(Y_i)} \leq n(C - 2\gamma') + n\gamma'\right) \\
&= P\left(\frac{1}{n} \sum_{i=1}^{n} \log \frac{P_{Y|X}(Y_i|X_i)}{P_Y(Y_i)} \leq C - \gamma'\right)
\end{aligned}$$

Since

$$\mathbb{E}\left[\log \frac{P_{Y|X}(Y_i|X_i)}{P_Y(Y_i)}\right] = I(X;Y) = C$$

for all i, we have that the probability above tends to zero by the weak law of large numbers. Clearly, the second term in the bound $2^{-\gamma} = 2^{-n\gamma'}$ also tends to zero because $\gamma' > 0$. So we have demonstrated a sequence of codes for which $\epsilon_n \to 0$ and the code rate is $C - 2\gamma'$ which is arbitrarily close to C.

c.) Now we again use the bound

$$\epsilon_n \le P\left(\log \frac{P_{Y^n|X^n}(Y^n|X^n)}{P_{Y^n}(Y^n)} \le \log M_n + n\gamma'\right) + 2^{-n\gamma'}.$$

with $\gamma' = \frac{\log n}{n}$ so the final term is $1/n$. Plug the value of M_n into the probability in the bound above. We have

$$\begin{aligned}
&P\left(\log \frac{P_{Y^n|X^n}(Y^n|X^n)}{P_{Y^n}(Y^n)} \le \log M_n + n\gamma'\right) \\
=&P\left(\sum_{i=1}^{n} \log \frac{P_{Y|X}(Y_i|X_i)}{P_Y(Y_i)} \le nC + \sqrt{nV}\Phi^{-1}(\epsilon) + \log n\right) \\
=&P\left(\frac{1}{\sqrt{nV}} \sum_{i=1}^{n} \left(\log \frac{P_{Y|X}(Y_i|X_i)}{P_Y(Y_i)} - C\right) \le \Phi^{-1}(\epsilon) + O\left(\frac{\log n}{\sqrt{nV}}\right)\right). \qquad (1)
\end{aligned}$$

Now note that for all $i \in [n]$,

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{\sqrt{V}}\left(\log \frac{P_{Y|X}(Y_i|X_i)}{P_Y(Y_i)} - C\right)\right] &= 0, \\
\mathrm{Var}\frac{1}{\sqrt{V}}\left(\log \frac{P_{Y|X}(Y_i|X_i)}{P_Y(Y_i)} - C\right) &= 1,
\end{aligned}$$

so the random variable in the probability converges to a Gaussian by the central limit theorem since the sum of n Gaussians is again Gaussian with $\mathcal{N}(\sum_i \mu_i, \sum_i \sigma_i^2)$, i.e., $\mu = 0$ and $\sigma^2 = n$. This is then normalized to a standard Gaussian by the remaining factor $1/\sqrt{n}$ in (1). Recall that for $X \sim \mathcal{N}(0,1)$

$$\Phi(x) := P\left(X \le x\right), \quad \text{s.t.} \quad P\left(X \le \Phi^{-1}(\epsilon)\right) = \Phi(\Phi^{-1}(\epsilon)) = \epsilon.$$

Consequently,

$$P\left(\log \frac{P_{Y^n|X^n}(Y^n|X^n)}{P_{Y^n}(Y^n)} \le \log M_n + n\gamma'\right) \to \epsilon$$

and we are done.

Exercise 7.4 *We consider the channel coding problem for a discrete channel with memory. The channel state $s \in \mathcal{S}$ is i.i.d. according to some pmf P_S and known to the decoder but not the encoder. The behaviour of the channel is then determined by a conditional pmf $W_{Y|XS}$. We are first interested in a one-shot converse bound. Consider an arbitrary code for sending M distinct messages over W. Such a code is given by an encoder $e : [M] \to \mathcal{X}$ and a decoder $d : \mathcal{Y} \times \mathcal{S} \to [M]$. As usual we consider the case where M follows a uniform distribution and let ε denote the average probability of error.*

a) *Derive the following bound, a meta-converse for this problem:*

$$|M| \le \max_{P_X} \min_{Q_{Y|S}} \frac{1}{\beta^*_\varepsilon(P_{XYS} \| P_X \times Q_{YS})}.$$

where $P_{XYS}(x,y,s) = P_X(x)P_S(s)W_{Y|XS}(y|x,s)$ and $Q_{YS}(y,s) = Q_{Y|S}(y|s)P_S(s)$.

b) *Further relax this bound and show that, for any $\delta \in (0, 1-\varepsilon)$,*

$$\log |M| \le \min_{Q_{Y|S}} \max_{x \in \mathcal{X}} D_s^{\epsilon+\delta}(P_{YS|X=x} \| Q_{YS}) + \log \frac{1}{\delta}.$$

c) *Consider the channel mutual information* $I(W) := \max_{P_X} I(X:Y|S)$ *with* P_{XYS} *as defined above. Show that*

$$I(W) = \min_{Q_{Y|S}} \max_{x \in \mathcal{X}} D(P_{YS|X=x} \| Q_{YS}) \,.$$

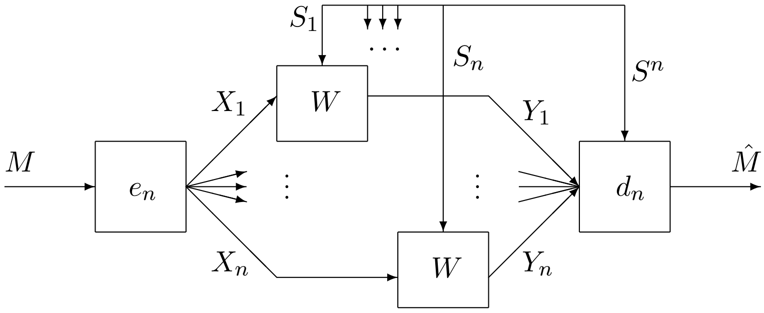


Figure 1: The channel coding setup for a fixed blocklength $n \in \mathbb{N}$.

Consider an arbitrary sequence of codes at rate R *sending* $M_n = \lceil 2^{nR} \rceil$ *messages over* W^n*. For each* n*, the code is given by an encoder* $e_n : [M_n] \to \mathcal{X}^n$ *and a decoder* $d_n : \mathcal{Y}^n \times \mathcal{S}^n \to [M_n]$*. As usual we consider the case where* $M \in [M_n]$ *follows a uniform distribution. Let* $\varepsilon_n = P[M \neq \hat{M}]$*. We want to show the strong converse, namely that for any such sequence of codes with* $\lim_{n\to\infty} \varepsilon_n < 1$*, we must have* $R \leq I(W)$*.*
Using the meta-converse in the form of

$$nR \leq \max_{x^n \in \mathcal{X}^n} D_s^{\epsilon+\delta}(W^n(\cdot|x^n) \| Q_Y^n(\cdot)) + \log \frac{1}{\delta},$$

we can conclude that for any sequence of codes with $\lim_{n\to\infty} \varepsilon_n < 1$*, there exists a* $\mu \in (0,1)$ *such that, for sufficiently large* n*, we have*

$$nR \leq \max_{x^n \in \mathcal{X}^n} D_s^{1-\mu}(P_{Y^n S^n | X^n = x^n} \| \hat{Q}_{YS}^{\times n}) + \log \frac{2}{\mu} \,,$$

where $\hat{Q}_{YS}(y,s) = P_S(s) \hat{Q}_{Y|S}(y|s)$ *with* $\hat{Q}_{Y|S}$ *the minimizer of Part c). Thus, in particular,* $D(P_{YS|X=x} \| \hat{Q}_{YS}) \leq I(W)$ *for all* $x \in \mathcal{X}$*.*

d) *Show that, for any* $\mu \in (0,1)$ *and any* $\nu > 0$*, and sufficiently large* n*,*

$$D_s^{1-\mu}(P_{Y^n S^n | X^n = x^n} \| \hat{Q}_{YS}^{\times n}) \leq n\,(I(W) + \nu)$$

for all $x^n \in \mathcal{X}^n$*. Complete the proof of the strong converse.*

Solution:

a.) Without of loss of generality, we consider the case where both the encoder and the decoder are deterministic. Consider the hypothesis testing problem with $H_0 = P_{XYS}$ and $H_1 = P_X \times Q_{YS}$ for some arbitrary channel $Q_{Y|S}$. Consider the test $\mathcal{A} = \{(x,y,s) \in \mathcal{X} \times \mathcal{Y} \times \mathcal{S} : x \neq e(d(y,s))\}$. The errors are

$$\alpha(\mathcal{A}) = P_{XYS}(\mathcal{A}) = P[X \neq e(d(Y,S))] = P[e(M) \neq e(\hat{M})] \leq P[M \neq \hat{M}] \leq \epsilon$$

and

$$\begin{aligned}
\beta(\mathcal{A}) = P_X \times Q_{YS}(\mathcal{A}^c) &= \sum_{xys\in\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}} P_X(x)Q_{YS}(ys)1\{x = e(d(y,s))\} \\
&= \sum_{m\in[M]} \frac{1}{|M|} \sum_{ys\in\mathcal{Y}\times\mathcal{Z}} Q_{YS}(ys)1\{e(m) = e(d(y,s))\} \\
&= \frac{1}{|M|} \sum_{ys\in\mathcal{Y}\times\mathcal{Z}} Q_{YS}(ys) \sum_{m\in[M]} 1\{e(m) = e(d(y,s))\} \\
&\le \frac{1}{|M|} \sum_{ys\in\mathcal{Y}\times\mathcal{Z}} Q_{YS}(ys) = \frac{1}{|M|}.
\end{aligned}$$

Since $\beta_\epsilon^\star(P_{XYS}\|P_X \times Q_{YS}) \le \beta(\mathcal{A})$ and the above arguments holds for any $Q_{Y|S}$ and the encoder e, we have that

$$|M| \le \min_{Q_{Y|S}} \frac{1}{\beta_\epsilon^\star(P_{XYS}\|P_X \times Q_{YS})}.$$

We may further maximise over P_X to get a bound that is valid for all encoders

$$|M| \le \max_{P_X} \min_{Q_{Y|S}} \frac{1}{\beta_\epsilon^\star(P_{XYS}\|P_X \times Q_{YS})}.$$

b.) Taking the logarithm of the above equation

$$\begin{aligned}
\log|M| &\le \max_{P_X} \min_{Q_{Y|S}} -\log\beta_\epsilon^\star(P_{XYS}\|P_X \times Q_{YS}) \\
&\le \max_{P_X} \min_{Q_{Y|S}} D_s^{\epsilon+\delta}(P_{XYS}\|P_X \times Q_{YS}) + \log\frac{1}{\delta} \\
&\le \max_{P_X} \min_{Q_{Y|S}} \max_{x\in\mathcal{X}} D_s^{\epsilon+\delta}(P_{YS|X=x}\|Q_{YS}) + \log\frac{1}{\delta} \\
&\le \min_{Q_{Y|S}} \max_{x\in\mathcal{X}} D_s^{\epsilon+\delta}(P_{YS|X=x}\|Q_{YS}) + \log\frac{1}{\delta},
\end{aligned}$$

where we have used the bound for asymmetric binary hypothesis testing and dropped the maximization over P_X since the expression is independent of P_X.

c.) For any P_X,

$$\begin{aligned}
&\min_{Q_{Y|S}} D(P_{XYZ}\|P_X \times Q_{YS}) = D(P_{XYS}\|P_X \times P_{YS}) + \min_{Q_{Y|S}} D(P_{YS}\|Q_{YS}) \\
&= D(P_{XYS}\|P_X \times P_{YS}) = H(X) - H(X|YS) = H(X|S) - H(X|YS) = I(X:Y|S),
\end{aligned}$$

where we have used that X is independent of S. Maximizing over P_X, we get

$$I(W) = \max_{P_X} I(X:Y|S) = \max_{P_X} \min_{Q_{Y|S}} D(P_{XYS}\|P_X \times Q_{YS}).$$

At the same time,

$$D(P_{XYS}\|P_X \times Q_{YS}) = \sum_{x\in\mathcal{X}} P_X(x) D(P_{YS|X=x}\|Q_{YS})$$

which is linear in $P(X)$ and convex in $Q(Y|S)$. We then use the minimax theorem to interchange the maximization and the minimization

$$I(W) = \max_{P_X} \min_{Q_{Y|S}} D(P_{XYS}\|P_X \times Q_{YS}) = \min_{Q_{Y|S}} \max_{P_X} D(P_{XYS}\|P_X \times Q_{YS}).$$

Note that the maximization over P_X is achieved by a single $x \in \mathcal{X}$ (for which then $P_X(x) = 1$)

$$\min_{Q_{Y|S}} \max_{P_X} D(P_{XYS} \| P_X \times Q_{YS}) = \min_{Q_{Y|S}} \max_{x \in \mathcal{X}} D(P_{YS|X=x} \| Q_{YS}).$$

This gives us

$$I(W) = \min_{Q_{Y|S}} \max_{x \in \mathcal{X}} D(P_{YS|X=x} \| Q_{YS}).$$

d.) We can find a bound on the information spectrum divergence

$$\begin{aligned} D_s^{1-\mu}(P_{Y^nS^n|X^n=x^n} \| \hat{Q}_{YS}^{\times n}) &= \sup\left\{K : P\left[\log \frac{P_{Y^nS^n|X^n=x^n}}{\prod_i \hat{Q}_{Y_iS_i}(Y_iS_i)} \le K\right] \le 1-\mu\right\} \\ &= \sup\left\{K : P\left[\sum_i \log \frac{P_{Y_iS_i|X_i=x_i}}{\hat{Q}_{Y_iS_i}} \le K\right] \le 1-\mu\right\}. \end{aligned}$$

The log-likelihood ratio is a sum of independent (but not identical) random variables, and thus it will concentrate around its expectation value

$$\mathbb{E}\left[\sum_i \log \frac{P_{Y_iS_i|X_i=x_i}}{\hat{Q}_{Y_iS_i}}\right] = \sum_i D(P_{Y_iS_i|X_i=x_i} \| \hat{Q}_{Y_iS_i}) \le nI(W)$$

where we have used the defining property for the last equation. By the law of large numbers, when n is sufficiently large, we have

$$P\left[\sum_i \log \frac{P_{Y_iS_i|X_i=x_i}}{\hat{Q}_{Y_iS_i}} \le nI(W)\right] \to 1.$$

Therefore, for any $\mu \in (0,1)$ and $\nu > 0$,

$$D_s^{1-\mu}(P_{Y^nS^n|X^n=x^n} \| \hat{Q}_{YS}^{\times n}) \le n(I(W) + \nu)$$

and thus combining with the inequality provided, for any such code, for sufficiently large n and arbitrarily small ν, we have

$$nR \le n(I(W) + \nu) + \log \frac{2}{\mu},$$

which is equivalent to

$$R \le I(W) + \nu + \frac{1}{n} \log \frac{2}{\mu}.$$

As ν can be arbitrarily small and n can be arbitrarily large, we must have

$$R \le I(W).$$